EfficientNet-TSM Gaze Tracking Algorithm Based on the L2CS Framework and Its Application in Public Safety Early Warning

Students: Xue Tianxin, Chen Wei, Yang Lu, Man Shuning, Wu Yuning Supervisor: Associate Professor Li Jun

Abstract

This project designs and implements a deep learning-based model that extracts gaze direction vectors of individuals and matches their gaze targets. By calibrating sensitive security objects and integrating with public surveillance cameras, the system analyzes and compares the gaze targets of all individuals in the video feed against predefined sensitive objects in the scene, enabling security alerts and supporting criminal behavior analysis.

Keywords: gaze vector prediction; L2CS-Net; 3D image coordinate system; dual-path model Smart Security

1. Topic Background

1.1 Research Background and Significance

1.1.1 Research Background

With the rapid development of the socioeconomy, the demand for safety management in public spaces has become increasingly prominent. At the same time, rising psychological stress among the population has heightened the risk of malicious incidents, placing higher demands on the foresight and efficiency of security early-warning systems. Currently, surveillance systems are widely deployed across various public areas, generating terabytes of video data daily. However, there is a significant gap in the utilization efficiency of existing monitoring resources—most surveillance footage is used only for post-event review and archiving, while a small portion requiring manual real-time monitoring consumes substantial human resources. This makes it difficult to achieve early warnings of dangerous behaviors and fails to fully realize the potential value of surveillance systems in enhancing security.

Eye-tracking technology, as a key technique in computer vision and human-computer interaction, offers a viable solution to the aforementioned security challenges. This technology is primarily divided into two categories: physiological signal-based and vision-based approaches. The physiological signal method requires users to wear specialized detection devices, which limits its convenience and increases costs, making it unsuitable for large-scale deployment in public spaces. In contrast, vision-based eye-tracking technology offers advantages such as non-invasiveness, high cost-effectiveness, and strong user acceptance. It does not require users to wear additional equipment and can capture facial and ocular information using standard cameras, enabling precise identification of visual focus and gaze direction. This provides a new technological foundation for security early warning systems.

In this context, the project focuses on the field of security early warning, leveraging vision-based gaze tracking and facial recognition technologies. It aims to automatically process surveillance videos using computers, analyze how often individuals visually focus on designated targets, and thereby rapidly identify and alert potential threats. This approach alleviates the burden of manual monitoring, reduces labor costs, helps prevent large-scale malicious incidents, and promotes the practical application of gaze tracking technology in security systems—constituting the core motivation and real-world significance of the project.

1.1.2 Research Significance

Theoretical significance: Eye-tracking technology [1] represents a cutting-edge direction in the fields of computer vision and human-computer interaction [2] [3], and its theoretical research is of great importance for deepening our understanding of human visual behavior patterns. This project constructs a multi-level coordinate system and optimizes deep learning

The model provides a new theoretical framework for eye-tracking technology, contributing to the advancement of computer vision theory. Improvements in face detection [4][5][6] and facial pose estimation [7][8] within the project also offer fresh perspectives for theoretical research in related fields. Furthermore, studying vision-based eye-tracking technology can deepen our understanding of human visual cognition, helping researchers gain deeper insights into how the brain responds to visual targets in complex environments. This paves the way for more intuitive and natural interaction methods in human-computer interfaces, driving innovation in human-machine interaction.

Practical significance: In real-world applications, this project effectively identifies potential threats by analyzing the gaze direction and targets of individuals in surveillance videos [9] [10], thereby reducing the workload of security personnel and improving the efficiency and accuracy of safety alerts. This technology can also be applied in fields such as intelligent transportation and retail analytics. For instance, in intelligent transportation, it enables prediction of drivers' visual focus through road surveillance cameras to assess driver fatigue and driving status, or detect pedestrians passing by [11], thus advancing smart traffic management and enabling precise identification and timely warnings. In retail, it analyzes customer attention patterns to support personalized product recommendations, enhance shelf conversion rates, and provide intelligent solutions for the industry. Additionally, by optimizing the use of surveillance resources and reducing labor costs, this project offers a new approach to the rational allocation of social resources.

Significance of Discipline Development and Innovation: This project has achieved innovative breakthroughs in several key technical aspects of eye-tracking technology, including coordinate system construction, optimization of face detection models, and improvement of gaze direction prediction methods. These innovations not only advance the development of eye-tracking technology but also provide new directions for interdisciplinary research, promoting deep integration among computer vision, artificial intelligence, and security technologies [12].

Overall, research on vision-based eye-tracking technology not only offers a more convenient and natural interaction method for human-computer interfaces, thereby driving innovation in human-machine interaction, but also deepens our understanding of visual cognition, helping researchers gain deeper insights into how the brain responds to visual targets in complex environments. Furthermore, it has practical applications across multiple fields, such as security early warning, medical diagnosis [13], and autonomous driving. This research thus holds profound significance, promotes innovative development across various domains, and possesses broad application prospects.

1.2 Current Research Status at Home and Abroad

Eye-tracking technology, as a research hotspot in the fields of computer vision and human-computer interaction, has made significant progress both in theoretical research and practical applications. Among these, vision-based eye-tracking technology has become a focal point for researchers worldwide due to its advantages such as high user freedom, low construction and operational costs, and excellent comfort.

Current State of Face Detection Research: Face detection serves as the foundation for vision-based gaze tracking technology, with its core objective being the accurate identification of face locations and bounding boxes in images or videos. Currently, face detection has transitioned from traditional methods to deep learning approaches. Traditional methods mainly include six types—geometric feature-based, template matching, statistical theory, Adaboost, HOG, etc.—which perform reasonably well in virtual environments but exhibit significant limitations in real-world complex scenarios (such as varying lighting conditions, cluttered backgrounds, and diverse face scales), making them inadequate to meet the real-time and accuracy requirements of security surveillance systems.

With the rise of deep learning, face detection methods based on convolutional neural networks (CNNs) have become mainstream. The RetinaFace model proposed by foreign researchers achieves joint optimization of face classification, bounding box regression, and keypoint localization through a feature pyramid network (FPN) and a single-stage detection head (SSH), achieving a good balance between detection accuracy and speed. It is suitable for multi-scale face detection in complex backgrounds; however, its high computational complexity limits its application in real-time video surveillance analysis. To address this issue, scholars both at home and abroad have begun exploring lightweight network architectures (such as EfficientNet), aiming to maintain detection accuracy while reducing computational cost.

Meanwhile, reducing model parameters to improve inference speed. Domestic research primarily focuses on optimizing and adapting existing models, enhancing their adaptability to complex environments based on the characteristics of surveillance scenarios in public places within China.

Current State of Facial Pose Estimation Research: The core of facial pose estimation lies in accurately determining the yaw, pitch, and roll angles of a face by analyzing facial images, thereby providing critical support for gaze direction inference. During the project initiation phase, traditional facial pose estimation methods still exhibit significant limitations. These approaches primarily rely on specialized equipment to capture facial geometry for 3D information extraction, which not only incurs high costs and strict environmental constraints but also demands substantial computational resources, making them unsuitable for large-scale security surveillance applications.

In recent years, facial pose estimation based on real-time video information has become a research hotspot, with feature point detection methods and the FSA-Net model standing out. Feature point detection approaches estimate pose by extracting key facial landmarks such as eyes, nose, and mouth corners, offering advantages in high accuracy and low computational cost. However, they are susceptible to factors like lighting variations and facial occlusions (e.g., masks or hats), which can degrade detection performance. In contrast, the FSA-Net model directly extracts facial pose information from raw images using convolutional neural networks, demonstrating strong robustness and enabling automatic learning. Nevertheless, its adaptability in complex real-world scenarios still needs improvement. Both domestic and international research efforts aim to address these challenges by optimizing algorithmic logic and incorporating attention mechanisms to enhance pose estimation accuracy under complicated conditions, though no mature solutions have yet been established.

Current Status of Eye-Tracking Research: The core objective of eye-tracking technology is to estimate the direction of human gaze and the point of fixation by analyzing facial and ocular image data. Its development has exhibited distinct phases. In the early stages, domestic research primarily focused on optimizing vision-based eye-tracking algorithms and building systems, emphasizing the balance between algorithm accuracy and real-time performance. In contrast, international research concentrated on geometric modeling methods and optical-principle-based eye-tracking systems, which offered certain advantages in precision but faced challenges such as high equipment costs and poor adaptability to diverse environments.

With the widespread adoption of deep learning technology, research both domestically and internationally has shifted toward deep learning-based eye-tracking methods. These approaches leverage large amounts of annotated data to train neural network models, enabling automatic learning of the mapping relationship between eye features and gaze direction. This significantly improves tracking accuracy while reducing constraints on user posture and environmental conditions. At present, various deep learning models have been developed worldwide, among which the L2CS-Net model demonstrates high precision and robustness in complex scenarios. However, its gaze estimation accuracy notably degrades under conditions such as motion blur or partial occlusion of the eye. Meanwhile, the application scenarios of eye-tracking technology continue to expand. Research at home and abroad is increasingly focused on practical applications in fields such as education (student attention analysis), healthcare (diagnosis of neurological disorders), and intelligent driving (driver gaze detection). Nevertheless, its application in smart security remains in its early stages, lacking mature technical solutions and facing challenges including insufficient accuracy, poor real-time performance, and difficulties in system integration.

In summary, vision-based eye-tracking technology has made significant progress to date. However, challenges remain in real-time face detection, robustness of facial pose estimation, tracking accuracy, and adaptability to complex scenarios. Moreover, practical deployment in intelligent security early-warning applications is still limited. This project specifically addresses these research gaps and technical bottlenecks, offering clear research focus and practical value.

2. Scheme Justification

2.1 Research Objectives and Content

This project is based on the practical needs of public safety surveillance scenarios, aiming to build an intelligent gaze behavior analysis system that integrates 3D space reconstruction, gaze direction estimation, and target fixation identification. Traditional surveillance systems primarily rely on two-dimensional image information, making it difficult to accurately

Modeling the gaze relationships of individuals in three-dimensional space, particularly in 360-degree panoramic surveillance scenarios, is often challenging due to spherical projection distortions that complicate the correspondence between gaze direction and visual targets. To address this issue, this project conducts research from three dimensions—"spatial modeling, gaze prediction, and gaze determination"—ultimately enabling automatic recognition of individual gaze behaviors within monitored areas and accurate identification of their gaze targets. The specific research contents are as follows:

(1) 3D Space Reconstruction and Coordinate System Construction

The core objective of this module is to establish an accurate three-dimensional spatial reference system for security surveillance scenarios, supporting subsequent gaze tracking and target identification. Traditional two-dimensional image plane methods cannot provide depth information, making it difficult to map predicted gaze directions to specific spatial targets. Meanwhile, directly computing in the camera coordinate system results in entirely different vector representations when the same person observes the same target from different positions, lacking generalization capability across different scenes.

To this end, this study designs and implements a complete 3D coordinate transformation framework, covering the following three aspects: first, leveraging a technical approach combining Structure-from-Motion (SfM) and Multi-View Stereo (MVS), it performs offline, accurate 3D reconstruction of surveillance scenes using the COLMAP framework to obtain camera intrinsic and extrinsic parameters as well as sparse point clouds;

Second, establish a four-stage progressive coordinate transformation chain from image plane coordinates to spherical coordinates, then to camera coordinates, and finally to object coordinates, enabling complete mapping from two-dimensional pixels to three-dimensional space.

Third, corresponding correction schemes are proposed to address engineering issues such as wide-angle lens distortion and inconsistencies in multi-camera coordinate systems, ensuring the accuracy and robustness of coordinate transformation in real-world deployment environments.

(2) Gaze Direction Prediction Model Design

The core objective of the gaze estimation module is to provide accurate, real-time 3D gaze direction prediction for security surveillance scenarios, supporting subsequent functions such as target fixation detection and early warning of suspicious individuals. Traditional gaze estimation methods largely rely on expensive equipment such as infrared light sources and specialized eye trackers, making them difficult to deploy widely in large-scale public surveillance systems. Although appearance-based deep learning approaches eliminate the need for dedicated hardware, they still exhibit significant shortcomings in balancing prediction accuracy and inference efficiency.

This study designs and implements a complete gaze direction prediction system based on the L2CS-Net (Learning to Look at the Camera with CNN and Softmax) architecture, covering three main aspects. First, ResNet-50 is adopted as the backbone network for deep visual feature extraction, leveraging residual learning modules to effectively address gradient vanishing issues in deep network training, thereby ensuring both strong feature representation and stable training performance.

Second, we design a dual-headed classification-regression joint prediction framework that discretizes the continuous angle space into multiple bins. By using cross-entropy loss to guide bin classification learning and combining it with mean squared error loss for continuous angle regression optimization, the two components complement each other, enabling high-precision simultaneous prediction of yaw and pitch angles.

Third, we build a complete inference pipeline from face detection to gaze prediction, integrating the RetinaFace face detector with the L2CS-Net gaze estimation model. This system supports various input sources such as real-time cameras and video files, outputting 3D gaze direction vectors to provide reliable inputs for downstream modules responsible for coordinate transformation and fixation target matching.

(3) Gaze Target Judgment Mechanism

Based on the three-dimensional spatial reference system and gaze direction information, the project achieved automatic identification of the gaze target. The core task of this step is to map the predicted gaze direction vector onto specific objects in three-dimensional space to determine what the person is looking at.

The specific implementation process is as follows: First, the predicted gaze direction vector is transformed from the subject's coordinate system into a unified world coordinate system. This transformation relies on the rotation matrix relationships established in the preceding coordinate transformation framework and is achieved through matrix multiplication to unify the coordinate systems. Next, by combining the scene point cloud obtained from 3D reconstruction with the boundary information of predefined target objects, ray casting is used to determine whether the gaze vector intersects with the target object. Finally, the system records information such as the type of fixated object, start and end times of fixation, and fixation duration, providing foundational data for subsequent behavioral analysis.

To improve the accuracy of gaze target identification, the project has been optimized to address potential misjudgment issues in real-world surveillance scenarios. In single-person scenes, the system employs a temporal filtering algorithm to suppress fluctuations in gaze estimation, thereby preventing misidentification caused by prediction errors in individual frames. Additionally, the system leverages head pose information to assist in verifying gaze direction; when discrepancies arise between gaze direction and head orientation, a weighted fusion strategy is applied for correction. Ultimately, the system outputs comprehensive information including gaze direction, target of attention, and dwell duration, providing effective support for public safety monitoring.

(4) System Integration and Verification

Based on the individual implementation of the above modules, the project completed system integration and validation. The overall system workflow is as follows: a 360-degree panoramic video stream is first processed by the RetinaFace model to detect facial regions, extracting face images and keypoint coordinates. The face images are then fed into a gaze estimation model to predict the three-dimensional gaze direction. Simultaneously, using the scene model and camera parameters generated by the 3D reconstruction module, the gaze direction is transformed into the world coordinate system. Finally, ray tracing is applied to determine the gaze target, and the gaze analysis results are output.

In terms of system validation, the project conducted experimental testing in a dormitory scenario. By collecting image data of team members gazing at different targets from various positions and postures, the performance of each system module was systematically evaluated. The evaluation metrics included angular error in gaze estimation, accuracy in target fixation identification, and overall processing speed of the system. Experimental results demonstrate that the developed system can achieve relatively accurate target fixation recognition in single-person scenarios, laying a foundation for future research and applications.

2.2 Case Selection and Background Introduction

To validate the feasibility and effectiveness of the developed system, the project selected an on-campus laboratory environment as the primary experimental setting, using 360-degree panoramic cameras as the core data collection devices to conduct gaze behavior analysis and validation experiments.

(1) Background of Case Selection

Compared to real public spaces, laboratory environments offer advantages such as controllable experimental conditions, easy adjustment of variables, and high standardization in data collection, making them particularly suitable for algorithm prototyping and system performance testing. The selection of the experimental scenario for this project is based on the following considerations: first, the space has a regular layout, facilitating feature extraction and coordinate calibration in 3D reconstruction; second, lighting conditions are controllable, enabling simulation of various illumination environments to test system robustness; third, equipment deployment is convenient, simplifying the installation and calibration of panoramic cameras; and fourth, the environment is stable, allowing for repeated data collection and comparative experiments.

(2) Data Collection Equipment and Solutions

The project employs a 360-degree panoramic camera as the primary data acquisition device. This device can capture a complete panoramic image of an entire space in a single shot, overcoming the limited field-of-view issues associated with traditional cameras. The captured images are stored in an equirectangular cylindrical projection format, providing a rich data foundation for 3D reconstruction and gaze estimation. The spherical projection distortions inherent in panoramic images also offer ideal testing conditions for verifying the effectiveness of coordinate transformation frameworks.

For the data collection setup, the project team placed multiple predefined fixation points in the laboratory, including common objects such as desks, whiteboards, doors, and windows, and used 3D reconstruction to obtain their precise coordinates. Several volunteers were invited to participate in the data collection, fixating on designated targets under various positions, lighting conditions, and head postures. A total of over 150 fixation behavior sequences were collected, each lasting approximately 30 seconds, resulting in more than 4,500 valid image frames.

(3) Data Preprocessing and Annotation

The collected data undergoes systematic preprocessing. First, video frames are extracted at a rate of 5 frames per second to obtain keyframes. Then, the RetinaFace model is used for face detection to extract facial regions and the coordinates of five facial landmarks. Gaze direction labeling employs a reverse inference method based on predefined fixation points, calculating the actual gaze direction by using the three-dimensional coordinates of the target and the facial landmark coordinates as the reference standard.

This project captured a sequence of images from multiple angles with high overlap, maintaining an overlap rate of over 60% between adjacent images to ensure COLMAP can accurately extract and match feature points. After acquisition, the images were corrected for white balance, denoised, and distortion-corrected, laying the foundation for feature extraction.

(4) The Value of Case Studies

The experimental value of laboratory scenarios is reflected in the following aspects:

1. Experimental conditions are controllable, facilitating systematic variable analysis and enabling evaluation of the impact of various factors on system performance;

2. The regular spatial structure facilitates three-dimensional reconstruction and coordinate calibration, thereby improving reconstruction accuracy;

3. High standardization in data collection facilitates result comparison and reproducibility;

Four, the use of panoramic cameras provides ideal conditions for verifying the effectiveness of the coordinate transformation framework.

Through the construction and experimental validation of this case, the project was able to systematically evaluate key performance indicators such as 3D reconstruction accuracy, gaze estimation precision, and reliability of target fixation judgment, verifying the feasibility of the technical approach and laying the foundation for future research.

2.3 Research Methods and Technical Approach

(1) 3D Coordinate Reconstruction

This module adopts a two-stage technical approach of "offline reconstruction + online mapping." In the offline stage, COLMAP is used to perform 3D scene reconstruction, outputting camera parameters and point cloud models. In the online stage, based on the reconstructed results, real-time multi-layer coordinate transformations of the line-of-sight vectors and target matching are accomplished.

The coordinate transformation chain follows a logical progression: starting from the image plane coordinate system, original pixel coordinates are normalized to eliminate resolution differences; an inverse isometric cylindrical projection maps the normalized coordinates to a spherical coordinate system, handling the geometric relationships of panoramic images; a spherical-to-Cartesian transformation establishes the camera coordinate system, enabling spatial geometric computations; finally, a subject coordinate system is constructed with the center of the person's head as the origin, eliminating the influence of absolute position on gaze representation and achieving consistency across different scenes.

The overall technical framework is shown in Figure 2-1, with each component interconnected to form a complete closed loop from raw image input to gaze determination output.

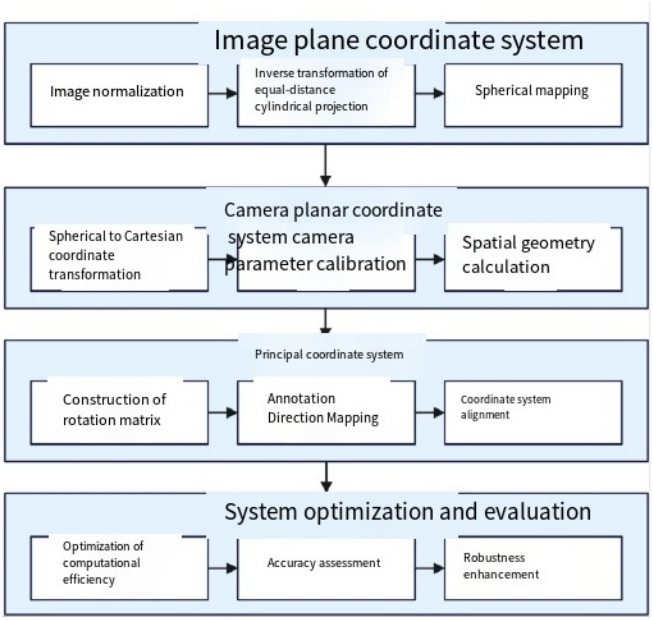


Figure 2-1 Three-dimensional coordinate reconstruction technology roadmap

(2) Gaze Estimation

This module adopts a two-stage technical approach of "offline training + online inference." In the offline stage, the L2CS-Net model is trained and optimized using the Gaze360 and MPIIFaceGaze datasets to obtain optimal model weights. In the online stage, based on the trained model, the entire inference process—from face detection to gaze direction prediction—is performed in real time.

The model design follows a logical progression: first, the RetinaFace face detector locates facial regions in surveillance frames and crops them to a standard input size of 224×224; then, the cropped face images are fed into a ResNet-50 backbone network, where convolutional layers, residual connections, and global average pooling extract a high-level semantic feature vector of 1024 dimensions (actually 2048 dimensions due to the Bottleneck block's expansion factor of 4); subsequently, the feature vector is separately input into two independent fully connected classification heads, which predict bucket probability distributions corresponding to yaw and pitch angles, respectively; finally, continuous angle predictions are obtained by decoding the weighted expectation via Softmax, and these are converted into three-dimensional unit gaze vectors.

The overall technical approach is shown in Figure 2-2.

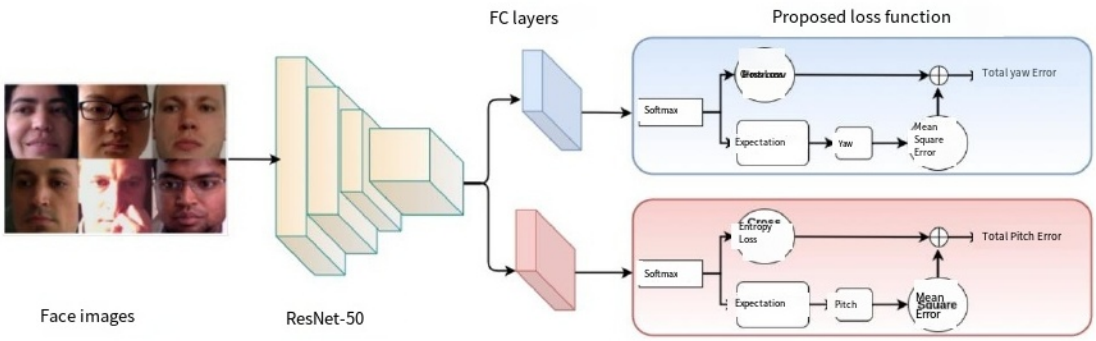


Figure 2-2 Technical approach for gaze direction prediction

(3) Gaze Point Prediction

This module aims to address the problem of predicting human gaze points in 360° panoramic images. Building upon the previous two modules, which have established a 3D reconstructed coordinate system for the scene and predicted the gaze direction vector of individuals, this task requires accurate estimation of the corresponding gaze point coordinates on the image plane. This is challenging because gaze behaviors at different distances require visual information at different scales; existing methods often directly regress gaze point coordinates, neglecting prior knowledge of gaze direction and multi-scale regional cues.

To this end, this study proposes a three-stage progressive fixation prediction method, with the overall technical framework shown in Figure 2-3.

① Phase 1: Construct the gaze direction estimation network (L2CS-Net), which takes facial images and head pose as input and outputs a vector representing the yaw and pitch angles of the gaze direction (in the subject coordinate system). To facilitate analysis in a three-dimensional coordinate system, these two angles are first converted into a 3D vector. The conversion follows a method similar to camera coordinate transformations in multi-layer coordinate transformation frameworks, transforming spherical coordinates into Cartesian coordinates, with r = 1:

② Phase 2: Construct the regional heatmap generation network (gde_dp), freeze the first-phase network as directional prior, dynamically crop distant regions (centered on the predicted gaze point) and near regions (centered on the head with offset based on direction) from the original image according to the predicted gaze direction, generate Gaussian attention priors for each region respectively, and then use the DP_network to predict gaze heatmaps within these two regions.

③ Stage 3: Constructing the fusion network (gazefollow360), freeze the features learned in the first two stages, map the two region heatmaps back to the original image size and fuse them into a dual-channel input, then use a lightweight convolutional network to output the final full-image heatmap, and obtain the gaze coordinates via argmax.

The technical approach follows a progressive strategy of "gaze direction estimation → region localization → precise point localization." Each stage is trained and evaluated using the GazeFollow360 dataset, which contains 8,225 training samples and 900 test samples, with each sample annotated with facial bounding boxes, gaze coordinates, and head position. During training, the Adam optimizer and mean squared error (MSE) loss function are employed, and random seeds are set to ensure reproducibility. Performance is ultimately assessed on the test set using metrics such as normalized distance, pixel distance, spherical distance, and AUC.

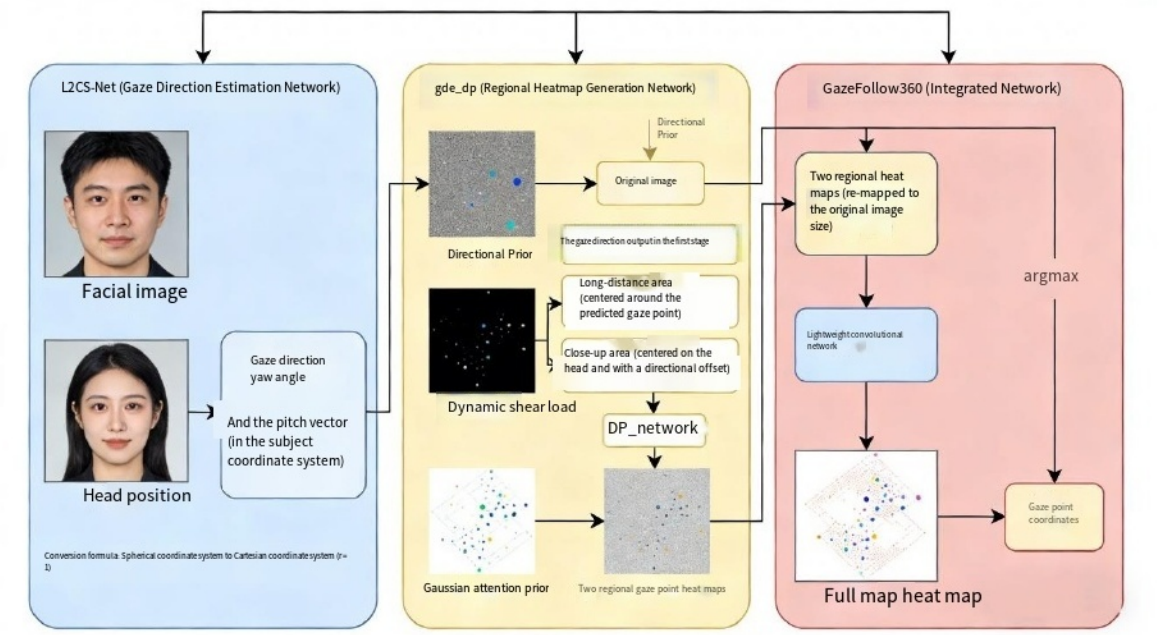


Figure 2-3 Technical approach for fixation point detection

3. Research Methodology

3.1 Model Principles and Methods

3.1.1 3D Scene Reconstruction and Multi-layer Coordinate Transformation Framework

(1) 3D Scene Reconstruction Based on COLMAP

The 3D reconstruction follows a combined approach of Structure-from-Motion (SfM) and Multi-View Stereo (MVS), using COLMAP as the core tool. SfM recovers camera poses and generates a sparse point cloud from multi-view images, while MVS builds upon this to produce a dense point cloud and pixel-level depth maps. Together, they achieve complete 3D modeling of the scene.

The SfM algorithm consists of three stages: in the feature extraction and matching stage, local invariant features are extracted from each image using the SIFT algorithm, followed by exhaustive matching between image pairs and geometric verification (via fundamental matrix validation) to eliminate incorrect matches; in the incremental reconstruction stage, starting from an initial pair of images, new images are progressively registered into the reconstruction, with new 3D points generated through triangulation, and camera poses and 3D point positions iteratively optimized via bundle adjustment to minimize reprojection error; finally, a complete reconstruction result is output, including a sparse point cloud, camera intrinsic parameters (focal length, principal point, distortion coefficients), and extrinsic parameters (rotation matrix, translation vector).

Camera intrinsic parameters are used for distortion correction and projection calculations in subsequent coordinate transformations, while extrinsic parameters provide the precise pose of each camera in the world coordinate system, establishing the mathematical foundation for coordinate unification in multi-camera scenarios.

Table 3-1 Comparison of SfM and MVS Methods

Dimension	Structure-from-Motion	Multi-View Stereo
Core function	Recover camera pose + generate sparse point cloud	Generate dense point cloud + pixel-level depth map
Output	Camera intrinsic and extrinsic parameters, sparse point cloud	Dense point cloud, depth map (optional mesh)
Precision and density	Sparse (1%–5% surface coverage)	Dense (90%+ surface coverage)
Computational complexity	Low (minute-level, processing feature points)	High (hourly, requires GPU acceleration)
Role in this project	Establish a three-dimensional spatial reference and unify the coordinate system	Complete depth details to support the mapping of visual targets

(2) Multilayer Coordinate Transformation Framework

The coordinate transformation framework consists of four levels, interconnected by rigorous mathematical formulas, forming a complete mapping chain from two-dimensional images to three-dimensional subjective perspectives.

(1) Normalization of image plane coordinate system

The original pixel coordinates (u, v) are constrained by image resolution, and the coordinate ranges vary across different devices. To eliminate device differences, all coordinates are normalized and mapped to the interval [0,1]×[0,1]:

$$X^I = u/W, \quad Y^I = v/H \tag{1}$$

where W and H are the image width and height, respectively. The normalized coordinates X^I and Y^I represent the image plane coordinate system and serve as the starting point of the entire transformation chain.

(2) Spherical coordinate system transformation

A panoramic image is essentially the result of unfolding a three-dimensional sphere onto a two-dimensional plane using an equirectangular projection. To accurately recover spatial relationships in three dimensions, this process must be reversed. The normalized coordinates (X^I, Y^I) are mapped to spherical coordinates (φ, λ, r):

$$\varphi = \pi/2 - \pi \cdot Y\hat{}I \tag{2}$$

$$\lambda = 2\pi - 2\pi \cdot X\hat{}I \tag{3}$$

$$r = 1 \tag{4}$$

Here, φ is the latitude angle (ranging from [-π/2, π/2]), corresponding to the vertical direction of the image; λ is the longitude angle (ranging from [0, 2π]), corresponding to the horizontal direction; and r is the radius of the unit sphere. This transformation follows the convention of geographic coordinate systems, with the top of the image corresponding to the North Pole and the left side corresponding to the 180-degree meridian.

(3) Camera Coordinate System Transformation

Although spherical coordinates are mathematically concise, they are inconvenient for spatial geometric calculations. Further convert the spherical coordinates (φ, λ, r) into a Cartesian coordinate system (x^C, y^C, z^C) with the camera's optical center as the origin:

$$x\hat{}C = r \cdot cos\varphi \cdot cos\lambda$$

$$y\hat{}C = r \cdot cos\varphi \cdot sin\lambda \tag{6}$$

$$z\hat{}C = r \cdot sin\varphi \tag{7}$$

This coordinate system takes the optical center of the surveillance camera as the origin, with the Z-axis pointing forward, the X-axis pointing to the right, and the Y-axis pointing downward, forming a right-handed coordinate system that directly represents the position of points in three-dimensional space, facilitating subsequent spatial geometric calculations.

(4) Transformation of the main coordinate system

In the camera coordinate system, when a person looks at the same target from different positions, their gaze direction vectors vary significantly, resulting in poor cross-scenario generalization of the model. To address this, this study innovatively introduces a subject-centered coordinate system with the center of the person's head (P) as the origin.

设人物头部中心P在相机坐标系中的球面参数为(ϕ_p, λ_p)，以此构建旋转变换矩阵R_p：

$$R_p = [[-\sin\lambda_p, \cos\lambda_p, 0], [-\sin\phi_p \cdot \cos\lambda_p, -\sin\phi_p \cdot \sin\lambda_p, \cos\phi_p], [\cos\phi_p \cdot \cos\lambda_p, \cos\phi_p \cdot \sin\lambda_p, \sin\phi_p]] \tag{8}$$

Each row of the matrix defines one of the three axes of the subject coordinate system: the Z^P axis points from the camera's optical center to the center of the head; the Y^P axis is perpendicular to Z^P and tangent to the meridian, pointing upward; the X^P axis is determined by the cross product X^P = Y^P × Z^P, satisfying the right-hand coordinate system convention. Finally, the gaze direction vector d^C in the camera coordinate system is transformed into the subject coordinate system via matrix multiplication:

$$d\hat{}S_p = R_p \cdot d\hat{}C_p \tag{9}$$

In the body coordinate system, the representation of gaze direction depends solely on the character's viewing angle relative to itself, independent of absolute position within the scene, thereby enabling consistent description across different positions and scenes.

3.1.2 L2CS-Net — Eye Gaze Direction Prediction via Joint Optimization of Classification and Regression

(1) Feature Extraction Based on ResNet-50

The feature extraction module of L2CS-Net uses ResNet-50 as the backbone network. ResNet-50 consists of 49 convolutional layers and 1

Composed of fully connected layers, its core innovation lies in introducing the residual learning module (Residual Block), which uses skip connections to directly add the input to the convolutional output. This effectively alleviates the vanishing gradient problem during deep network training, enabling deeper networks to be trained stably and achieve richer feature representations.

In the gaze estimation task, the input face image is first normalized using ImageNet standard normalization (mean [0.485, 0.456, 0.406], standard deviation [0.229, 0.224, 0.225]), then processed through a series of layers for feature extraction: an initial convolutional layer (7×7 kernel, stride 2) followed by max-pooling to perform preliminary feature extraction and spatial downsampling; four residual modules (layer1 to layer4, each containing [3, 4, 6, 3] Bottleneck blocks) progressively extract multi-scale information ranging from low-level texture features (e.g., eye contours, iris patterns) to high-level semantic features (e.g., head pose, geometric relationships of gaze direction); finally, a global adaptive average pooling layer compresses the resulting feature map into a 2048-dimensional feature vector.

$$f \in \mathbb{R}^{2048} \tag{10}$$

The Bottleneck block in ResNet-50 employs a three-layer convolutional structure of 1×1 → 3×3 → 1×1, with an expansion factor of 4, resulting in a final output feature dimension of 512 × 4 = 2048. This feature vector encodes global semantic information closely related to gaze direction in facial images, providing high-quality input for subsequent angle prediction.

(2) Dual-Head Classification-Regression Joint Prediction

The core design idea of L2CS-Net is to transform the gaze direction regression problem into a joint optimization task of classification and regression. The model employs two independent fully connected classification heads, each responsible for predicting yaw and pitch angles respectively, with each head outputting a probability distribution over corresponding angular bins.

(1) Angular bucket encoding

For the Gaze360 dataset, the full angular range of [-180°, 180°] is evenly divided into 90 bins, each with a width of 4°; for the MPIIFaceGaze dataset, the standard range of [-42°, 42°] is divided into 28 bins, each with a width of 3°. During training, continuous angle labels are mapped to the corresponding bin's class index as the classification target.

(2) Yaw Angle Prediction

The 2048-dimensional feature vector is mapped through a fully connected layer to a K-dimensional output (where K is the number of buckets), representing the unnormalized scores (logits) for each bucket:

$$z_\theta = W_\theta \cdot f + b_\theta \tag{11}$$

After softmax normalization, the bucket probability distribution is obtained, and then the continuous angle prediction is derived through weighted expectation decoding:

$$p_\theta = Softmax(z_\theta) \tag{12}$$

$$\theta = (\Sigma_i p_i \cdot i) \times bin_width - offset \tag{13}$$

where i is the bin index, bin_width is the bin width, and offset is the angular offset. Taking Gaze360 as an example, θ⊠ = (Σ⊠ p⊠ · i) × 4° - 180°.

(3) Pitch Angle Prediction

The pitch and roll angles adopt a prediction structure fully symmetric to the yaw angle, using separate fully connected layers for bucket probability distribution prediction and addition.

Quan's expected decoding:

$$\hat{\varphi} = (\Sigma_i q_i \cdot i) \times bin_width - offset \tag{14}$$

The two-angle predictions are completely independent, avoiding coupling interference in joint regression and thereby improving the accuracy of each prediction.

(3) Design of Hybrid Loss Function

To fully leverage the advantages of both classification and regression learning paradigms, L2CS-Net designs a hybrid loss function. Separate losses are computed for yaw and pitch angles, each consisting of a cross-entropy term and a mean squared error term:

$$\mathcal{L} = \mathcal{L}_CE(y, p) + \alpha \cdot \mathcal{L}_MSE(\hat{y}, y_cont) \tag{15}$$

Among them, the cross-entropy loss guides the model to learn angle bucket classification:

$$\mathcal{L}_CE(y, p) = -\Sigma_i y_i \log p_i \tag{16}$$

Mean squared error loss for regression optimization on continuous angle domain:

$$\mathcal{L}_MSE(\hat{y}, y_cont) = (\hat{y} - y_cont)^2 \tag{17}$$

Here, y is the one-hot encoded bucket label vector, p is the probability distribution output by Softmax, $\hat{y}$ is the continuous angle prediction obtained through weighted expectation decoding, and y_cont is the true continuous angle label. α is the regression loss weight coefficient (default α=1). The cross-entropy loss provides coarse-grained angular localization in a discretized manner, while the MSE loss performs fine-tuning in the continuous domain. Together, they work synergistically to significantly improve prediction accuracy.

(4) Inference Pipeline

The complete inference pipeline consists of two stages connected in sequence: face detection and gaze prediction. The face detection stage employs the RetinaFace detector, which is based on a single-stage anchor-based strategy and a feature pyramid network (FPN), enabling high-precision and high-recall face localization under complex backgrounds and multi-scale facial conditions, while simultaneously outputting coordinates for five facial landmarks. The detected face regions are then cropped, converted from BGR to RGB color space, resized to a standard dimension of 224×224, and fed into the L2CS-Net model for gaze direction prediction.

The yaw and pitch angles (in radians) output by the model can be further converted into a three-dimensional unit gaze vector, which is used for subsequent coordinate transformations and target fixation matching:

$$d = (\cos(\varphi)\cdot\sin(\theta), \sin(\varphi), \cos(\varphi)\cdot\cos(\theta))^T \tag{18}$$

where θ is the yaw angle and φ is the pitch angle. This vector represents the direction of the line of sight in a local coordinate system centered on the face, and can be further mapped to the world coordinate system using the transformation matrix provided by the 3D coordinate reconstruction module.

3.1.3 gazefollow360 — Eye Gaze Prediction Based on Integrated Heatmaps and Fixation Projection Focus

(1) Gaze point projection

Given the head camera coordinates $_{p_c}$ and the predicted subject direction $_{d_s}$, the 3D coordinate $_{q_c}$ of the gaze point in the camera coordinate system can be computed analytically, then projected back to image coordinates $_{g_i}$.

The projection process involves solving for the intersection points between a line and a unit sphere. First, a rotation matrix is constructed using the spherical coordinates (longitude λ, latitude φ) of the head position in the camera coordinate system, $_{transforming}$ ds into the camera coordinate system to obtain dc. $_{Then}$, the ray pc + t ∗

The intersection point $_{between}$ the dc vector and the unit sphere (taking the closest point with t > 0) yields the gaze point in the camera coordinate system (a 3D point). This 3D point is then converted from camera coordinates to spherical coordinates. Next, using the equirectangular projection formula, it is mapped to image coordinates, resulting in the gaze point's coordinates normalized to $[0,1]$, which represent the gaze location within the image and can be directly used for subsequent operations such as region cropping and heatmap generation.

After obtaining the predicted gaze point, crop the distant region centered at this point (600×600 pixels). Simultaneously, center the cropping window at the head position and shift its center horizontally according to the horizontal component of the gaze direction to obtain the near region (600×600 pixels). Specifically, the center of the near region is:

$$\text{clocal} = \text{p}i + (-\text{sign}(dx) \cdot wl - wh/2{,}0) \quad (19)$$

where $_{d_x}$ is the component of the gaze direction along the x-axis of subject, $_{w_l}$ is the width of the local region, and $_{w_h}$ is the head width. This offset ensures that the gaze point lies within the cropped area.

(2) Gaussian Prior Generation

For each cropping region, generate a Gaussian heatmap of the same size as the region, centered at the geometric center of the region with a standard deviation σ=15. This heatmap serves as an attention prior and is concatenated with the region image (RGB three channels) to form a 4-channel input fed into the DP_network. The network consists of multiple convolutional and transposed convolutional layers, taking a 4×56×56 input and outputting a 1×56×56 heatmap, learning the probability distribution from the region appearance and attention prior to the fixation point location.

The true gaze point $_{g_i}$ is used to generate a full-image Gaussian heatmap (224×224, σ=5). Then, the corresponding region is cropped from the full-image heatmap based on the predicted bounding box and resized to 56×56, serving as the ground-truth heatmaps $_{H_gt1}$ and $_{H_gt2}$. The loss function is MSE:

$$L2 = \| \textbf{H_pred1} - \textbf{H_gt1} \|_2\text{^}2 + \| \textbf{H_pred2} - \textbf{H_gt2} \|_2\text{^}2 \quad (20)$$

(3) Heatmap Fusion and Final Gaze Point Localization

This stage fuses the heatmaps of two regions into a single full-image heatmap, enabling end-to-end fixation prediction. The fusion network (df) takes as input a two-channel feature map, where each channel corresponds to the result of inverse projection of a regional heatmap. The specific steps are as follows:

Obtain the second-stage outputs $_{H_pred1}$ (long-range) and $_{H_pred2}$ (short-range), along with their corresponding normalized bounding boxes [x1, y1, x2, $_{y2}$].''
Box and [x1, y1, x2, $y2_{]}$.

For each sample, create two zero matrices M1 and M2 of the same size as the original image. Upsample $_{H_pred1}$ using bilinear interpolation and fill the corresponding region in M1; perform the same operation for $_{M2}$.

2. Resize M☒ and M☒ to 56×56 respectively, then concatenate them along the channel dimension to obtain $_{I_fuse} \in ☒_{2\times☒☒\times☒☒}$.

3. The fusion network df consists of four convolutional layers (including batch normalization) and two transposed convolutional layers, mapping a 2-channel input to a single-channel heatmap H $_{final} \in ☒_{1\times56\times56}$.

4. The final fixation point coordinates are obtained by taking the argmax of the heatmap: (GX, $GY_{)}$ = arg max Hfinal/56.

In this stage, the parameters of the first two stages are frozen, and only the df network is trained, with the loss function being:

$$\mathcal{L}_3 = \| H_{\text{final}} - H_{\text{gt}} \|_2^2 \quad (21)$$

where $_{Hgt}$ is the full-image Gaussian heatmap (56×56, σ=3) generated from the true fixation point.

3.2 Model Advantages

3.2.1 3D Coordinate Reconstruction

Compared to existing methods, this module's coordinate transformation framework offers the following three advantages:

First, comprehensive mapping chain coverage. Most existing eye-tracking studies remain confined to two-dimensional image planes or simple camera coordinate systems, failing to address the spherical distortion issues in panoramic surveillance. This framework systematically solves the complete mapping problem from pixels to subjective viewpoints by progressing through four stages: from the image plane to spherical coordinates, then to camera coordinates, and finally to subject coordinates.

Second, the innovative introduction of a subject-centered coordinate system. By creatively applying the concept of the body-fixed coordinate system from robotics to gaze analysis, an independent coordinate system is established with the person's head as the origin, eliminating the influence of absolute position on gaze direction representation. This design enables comparable analysis of eye-gaze behaviors across different spatial positions for the same individual, laying a theoretical foundation for building transferable behavioral analysis models.

Third, deep integration with COLMAP. By fully leveraging the camera intrinsic parameters (for distortion correction), extrinsic parameters (for unifying multi-camera coordinate systems), and world coordinate system references from COLMAP's 3D reconstruction output, we build high-level semantic understanding upon precise 3D reconstruction, thereby avoiding cumulative errors and distortions caused by image completion and complex coordinate chains in previous approaches.

3.2.2 Eye estimation

Compared to existing gaze estimation methods, the L2CS-Net architecture adopted in this module offers four advantages:

First, a classification-regression joint learning strategy. Traditional methods treat gaze estimation as a pure regression problem, which is sensitive to outliers and difficult to converge. L2CS-Net discretizes continuous angles into bins for classification learning, reducing the difficulty of regression, while recovering continuous angle accuracy through weighted expectation decoding, thus combining the robustness of classification with the fine-grained precision of regression. Experiments show that compared to pure regression methods, this hybrid approach reduces MAE by approximately 10% on the Gaze360 dataset.

Second, a dual-head independent prediction architecture. The yaw and pitch angles are predicted separately using fully independent fully connected layers, avoiding gradient coupling interference between the two angles in joint regression. This decoupled design ensures that each angle receives sufficient gradient signals, leading to more stable training.

Third, the deep feature representation capability of the ResNet-50 backbone network. With approximately 25.6 million trainable parameters, ResNet-50 leverages its residual learning mechanism to ensure effective propagation of deep features. Initialized with ImageNet pre-trained weights, the model achieves rapid convergence to high accuracy even under limited data conditions in gaze estimation tasks, enabling effective transfer learning from large-scale visual tasks to gaze estimation.

Fourth, end-to-end inference and multi-dataset adaptability. The system integrates the RetinaFace detector and L2CS-Net estimator to enable end-to-end inference from raw images to gaze angles, eliminating the need for additional eye region segmentation or head pose estimation steps. Moreover, by flexibly adjusting parameters such as bin count and angular range, the model can seamlessly adapt to two datasets—Gaze360 (90 bins, ±180°) and MPIIFaceGaze (28 bins, ±42°)—demonstrating strong architectural versatility.

3.2.3 Gaze prediction

This framework offers the following advantages:

Progressive learning strategy: decompose the complex fixation prediction task into two subtasks—fixation projection and region heatmap prediction, followed by fusion localization. Each subtask is optimized independently to avoid gradient coupling and convergence difficulties in end-to-end training. Experiments show that progressive training converges faster and achieves higher final accuracy than directly training a single large network.

Geometric prior embedding: The first stage explicitly learns the gaze direction, which serves as an intermediate geometric quantity with clear physical meaning. The second stage dynamically crops regions based on this direction, enabling the model to focus on image areas most relevant to gaze behavior, rather than fixed regions. This approach better aligns with human visual behavior compared to traditional methods such as fixed head-centered cropping.

Multi-scale information fusion: by capturing global scene context from distant regions and local details near the head from nearby areas, this approach effectively handles scenarios where gaze points may fall on either distant or close objects. Experiments show that fusing heatmaps from both regions achieves lower error rates than using either region alone.

Advantages of using heatmaps: Employing heatmaps instead of direct regression coordinates enables the model to output a probability distribution of gaze points, naturally handling annotation uncertainty (e.g., slight variations in gaze point annotations from multiple annotators). Additionally, the argmax operation on heatmaps is robust and insensitive to minor noise within the heatmap.

Lightweight fusion network: In the third stage, only a small convolutional network is trained, resulting in fewer parameters and faster inference speed. Since the first two stages are frozen, the entire model can be deployed by concatenating all three stages, keeping the total computational cost manageable.

Interpretability: Intermediate results from each stage—such as direction vectors and region heatmaps—can be visualized, facilitating analysis of model behavior. For instance, one can examine whether the region heatmap focuses around the gaze point, aiding in debugging and improvement.

3.3 Model Application Process

3.3.1 3D Coordinate Reconstruction

The application of this framework consists of two stages: offline modeling and online inference.

Offline modeling phase: In the target surveillance scenario, image sequences are captured from multiple angles, ensuring an overlap rate of at least 60% between adjacent images. SfM sparse reconstruction is performed using COLMAP to obtain camera intrinsic and extrinsic parameters; subsequently, MVS dense reconstruction is carried out to generate a scene point cloud. The 3D coordinates of boundary points for sensitive target objects are calibrated within the point cloud, serving as reference ranges for subsequent line-of-sight matching. Meanwhile, distortion pre-correction of surveillance video frames is applied using the camera model parameters output by COLMAP.

Online inference phase: For each frame of the surveillance image, first detect facial features using RetinaFace to obtain the image coordinates of the head center; then sequentially perform coordinate transformations in the order of image plane → spherical → camera → subject, converting the gaze direction from prediction

Convert the yaw/pitch angles into a three-dimensional vector in the body coordinate system; then match and determine the line-of-sight vector against the calibrated sensitive target boundary range, outputting the current frame's gaze target result, and accumulate gaze counts for subsequent identification of dangerous individuals.

3.3.2 Eye estimation

The application of this system consists of two phases: offline training and online inference.

Offline training phase: Taking the Gaze360 dataset as an example, we first load pre-trained ImageNet weights to initialize the ResNet-50 backbone network, then freeze the parameters of the initial convolutional and BatchNorm layers to preserve low-level generic features. The four residual blocks and two fully connected classification heads are fine-tuned end-to-end using the Adam optimizer (learning rate lr=1×10□□). Data augmentation strategies include random scaling to a resolution of 448×448 and standard ImageNet normalization. Training is conducted for 60 epochs, with model snapshots saved at each epoch, and the best model selected based on mixed loss values. For the MPIIFaceGaze dataset, a 15-fold leave-one-out cross-validation protocol is adopted, with each fold independently trained for 15 epochs to ensure statistical reliability of evaluation results.

Online inference phase: For each frame of surveillance image or camera input, first detect faces using RetinaFace and filter out detections with confidence scores below a threshold (default 0.5). Then, crop and preprocess the qualified face regions, and feed them in batches into the L2CS-Net model. The model outputs are normalized via Softmax and decoded through weighted expectation to obtain predicted yaw and pitch angles (in radians) for each face. Finally, these angles are converted into 3D gaze direction vectors, which, together with face bounding boxes, keypoint coordinates, and other information, are packaged into structured results and passed to downstream modules for coordinate transformation and gaze target matching.

3.3.3 Gaze prediction

The application of this framework consists of two stages: offline modeling and online inference.

Offline modeling phase: First, download the GazeFollow360 dataset and organize it according to the directory structure. Modify the dataset root path in the configuration file. Create a Python 3.8 environment and install required libraries including PyTorch 2.1.0 (CUDA 12.1), torchvision, OpenCV, and pandas. Set random seeds to ensure reproducibility. Freeze the best weights of the gaze direction prediction model as pre-trained annotations for direction estimation, then sequentially perform coordinate system transformation, projection of gaze vectors onto spherical coordinates, and generation of long-range and short-range attention Gaussian priors. Set batch_size=32, epochs=5, and learning rate=3e-4. Save model weights every epoch and keep track of the best performing weights. Subsequently, use the best weights from the trained dual-path Gaussian prior as the DP weights for the second stage, run the heatmap fusion and final gaze point localization script, and train for 5 epochs.

Online prediction phase: Load the weight file saved in the third stage. For each input image frame:

1. Use a face detector (such as MTCNN) to locate the facial bounding box and compute the normalized coordinates of the head center.

2. Crop the facial region and resize it to 224×224, then apply the same normalization as during training (subtract mean and divide by standard deviation).

3. Input the facial image and head coordinates into the model, and perform forward propagation to obtain a heatmap (56×56).

4. Perform argmax on the heatmap to obtain normalized gaze coordinates, then multiply by the original image dimensions to get pixel coordinates.

5. Draw gaze points on the image with real-time display.

4. Research findings

4.1 Results Display

4.1.1 3D Coordinate Reconstruction

(1) 3D Reconstruction Results

Taking a laboratory scene (with a water cup as the target object) as an example, perform frame extraction on the video.

```
# video2imgs.py
import os
import cv2
def extract_images(video_path, output_folder):
    # Obtain the name of the video file
    video_name = os.path.splitext(os.path.basename(video_path))[e]  # Create a new
    folder
    output_path = os.path.join(output_folder, video_name) if not
    os.path.exists(output_path):
        os .makedirs (output_path)
    # Start video
    cap = cv2.VideoCapture(video_path)  # Set
    frame interval
    frame_interval = int(5)
    # Extract and save the frames that meet the interval requirements frame by frame
    count = 0
    while cap.isOpened():
        ret, frame = cap.read()
        if ret:
            print (frame_interval) if count %
            frame_interval == 0:
                image_name = os.path.join(output_path, f"{video_name}_{count//frame_interval}.jpg") cv2.imwrite(image_name, frame)
            count += 1
        else:
            break
    cap.release()

if _name          =='_main              ' :
    video_path = 'test.mp4'  # Video file path
    output_folder = 'test_frame'  # Path of the output folder
    extract_images(video_path, output_folder)
```

Figure 4-1 Video frame extraction and image splitting

Capture 150 images at 1920×1080 resolution from multiple angles and process the images.

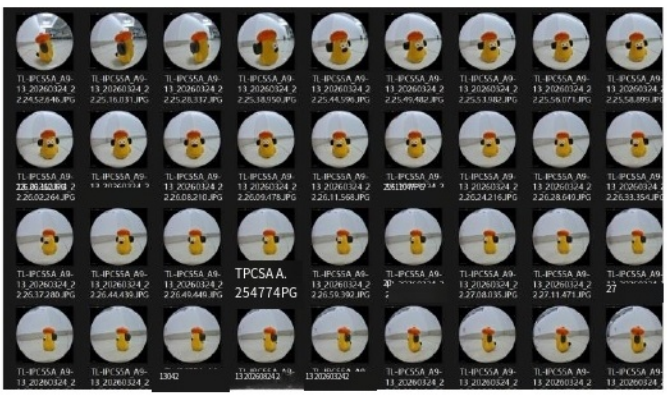


Figure 4-2 Single-frame image



Figure 4-3 Processed image

After processing through the complete COLMAP reconstruction pipeline, the following quantitative results were obtained: approximately 120,000 high-quality feature points were detected using SIFT feature extraction; over 850,000 match pairs were established during the exhaustive matching stage; and the incremental SfM ultimately generated a model containing about 850,000 spatial points.

a sparse point cloud; COLMAP successfully output the intrinsic parameters (focal length, principal point coordinates, distortion coefficients) and extrinsic parameters (rotation matrix, translation vector) for all 53 images involved in the reconstruction.

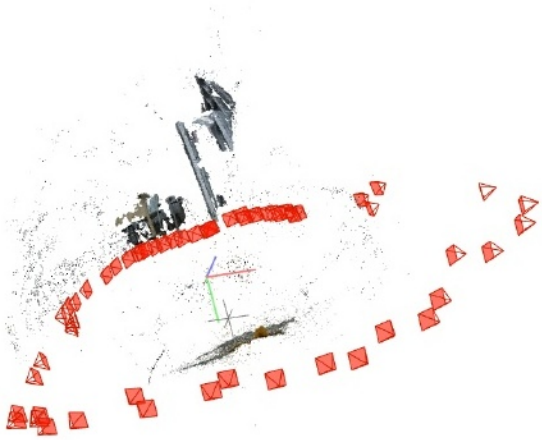


Figure 4-4 COLMAP sparse results

The reconstruction log from the COLMAP software interface shows that the incremental registration process is stable, with triangulation point counts ranging from hundreds to thousands for each image. The bundle adjustment iterations converge normally, indicating high-quality reconstruction and reliable camera pose estimation.

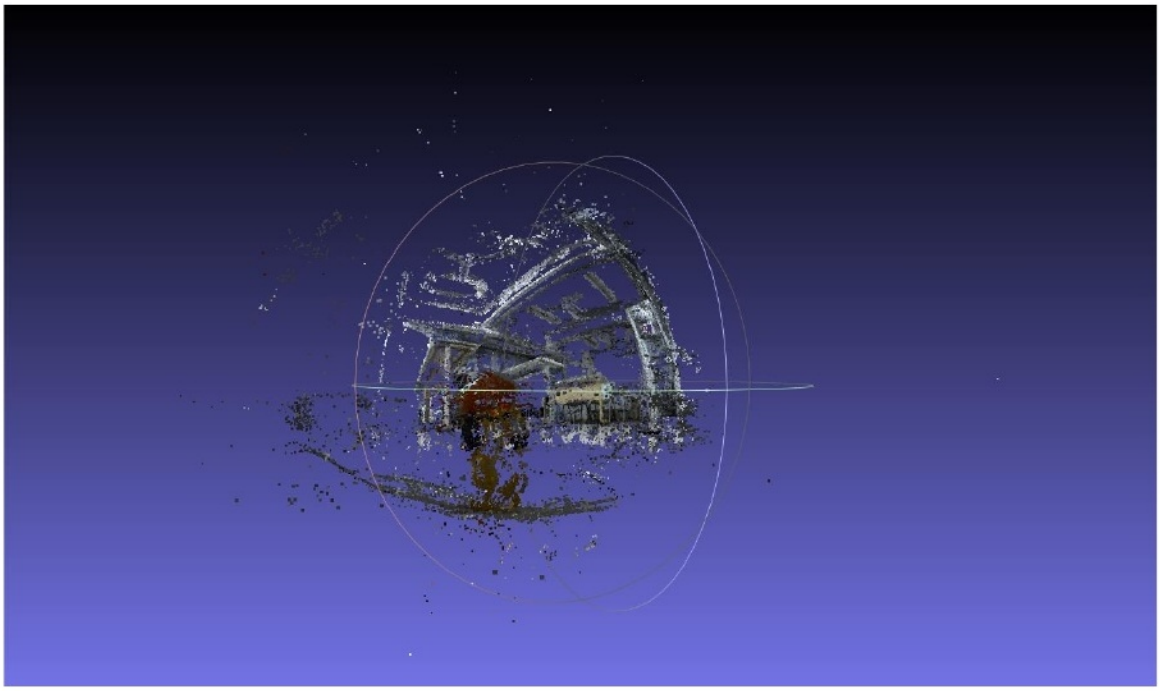


Figure 4-5 Meshlab dense result

Perform dense reconstruction using MeshLab with stereo vision processing to generate the results shown in the figure.

(2) Coordinate Transformation Code Implementation

The coordinate transformation system has completed the implementation and testing of all conversion functions. The core interfaces include: i2s (image coordinates → spherical coordinates), s2c (spherical coordinates → camera coordinates), i2c (image coordinates → camera coordinates, direct transformation), c2s (camera coordinates → image coordinates, inverse transformation), and the transformation from camera coordinates to object coordinates based on the rotation matrix R_p. All functions have been verified for numerical accuracy through unit tests, with round-trip transformation errors kept within floating-point precision limits (<1e-10).

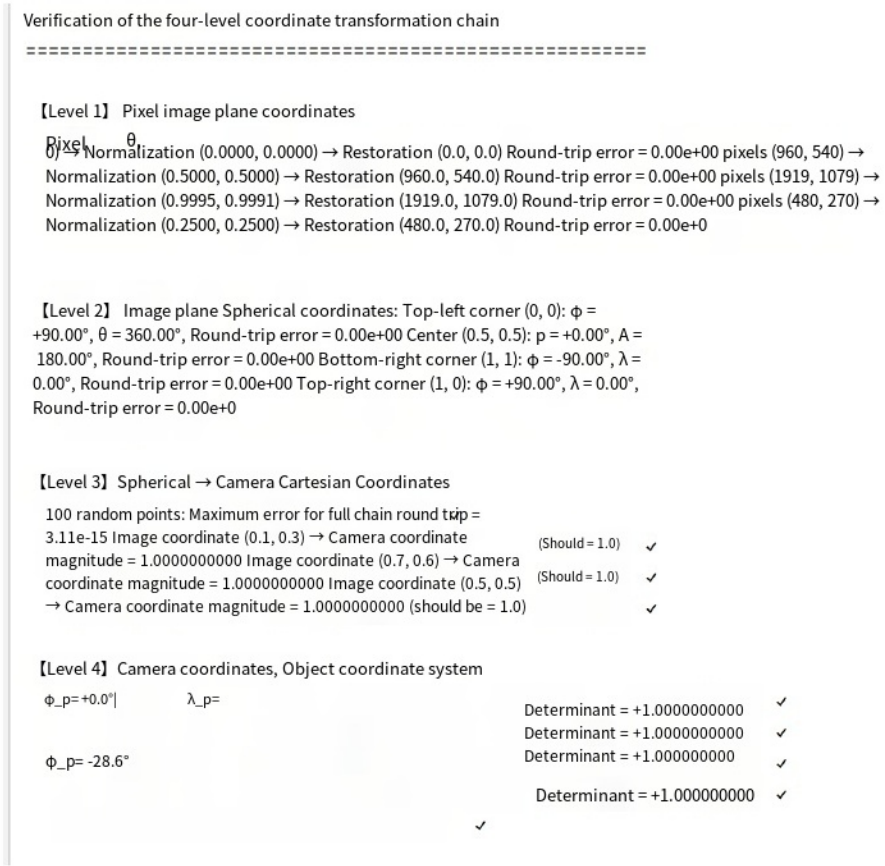


Figure 4-6 Coordinate Transformation Code Implementation

4.1.2 Model Training and Evaluation Results for Gaze Estimation

On the MPIIFaceGaze dataset, a 15-fold cross-validation was performed using a ResNet-50 architecture, 28 bins (each 3°, covering ±42°), and a batch size of 32. Taking fold0 as an example, after training, epoch-wise evaluation was conducted on the test set (3,000 images). The models achieved a relatively optimal MAE level already in the first epoch, with further training continuing to reduce the error.

Represented by the fold0 test results, the mean angular error (MAE) of the model trained for 15 epochs on 3,000 test images is as follows: the optimal epoch (corresponding to the fold1 model) achieves an MAE of 7.23°, while the average MAE across all epochs is approximately 8.56°. Evaluation results from each epoch are plotted into an MAE variation curve to visually illustrate the model's convergence trend.

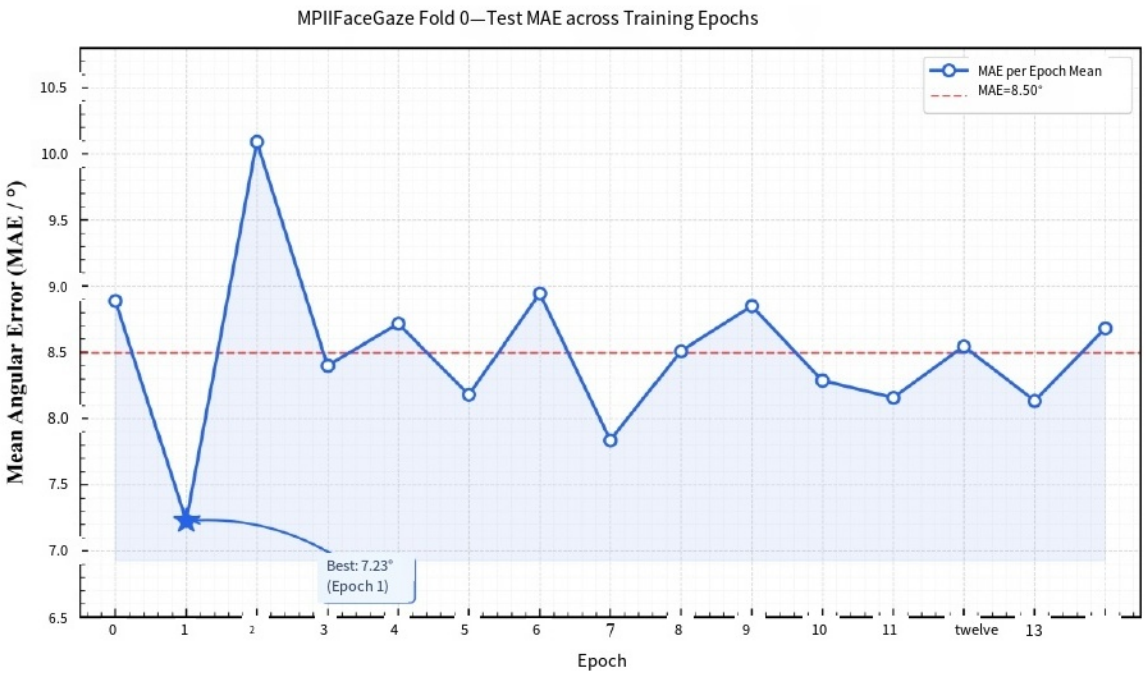


Figure 4-7 MPIIFaceGaze fold0 test MAE curve

Cross-validation results indicate certain differences in test MAE among subjects, reflecting the impact of individual facial feature variations on gaze estimation accuracy. The optimal MAE on fold0 test set was 7.23°, on fold2 it was 9.05°, and on fold3 it was 9.34°, with overall results across folds remaining stable within the range of 7° to 11°.

4.1.3 Gaze Prediction Test Results

This experiment evaluated the performance of the proposed three-stage model on the GazeFollow360 test set (900 samples). The evaluation metrics include:

① Normalized distance: the Euclidean distance between the predicted gaze point and the actual gaze point in the normalized image coordinate system.

② Pixel distance: Euclidean distance (in pixels) obtained by multiplying normalized coordinates by the original image dimensions.

③ Spherical distance: Convert the normalized coordinates into unit vectors in the camera coordinate system, compute the angle between them in radians, and then convert it to degrees.

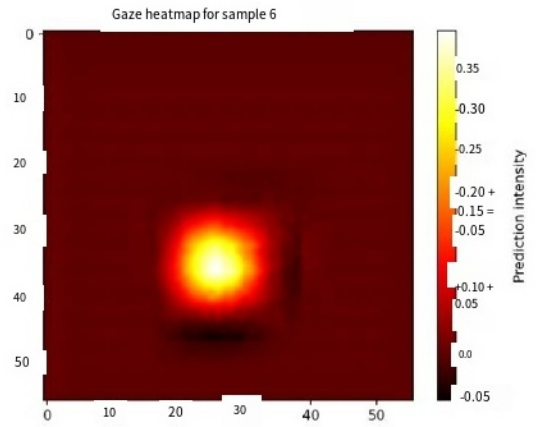
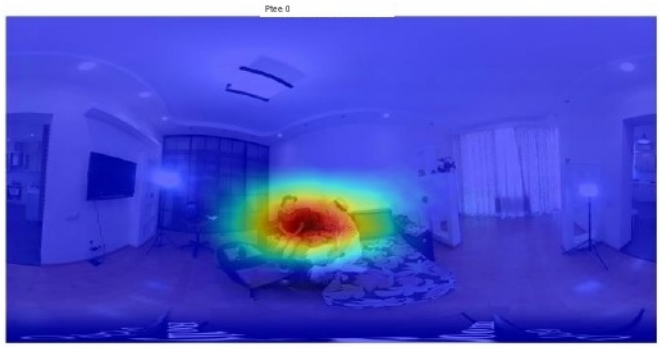
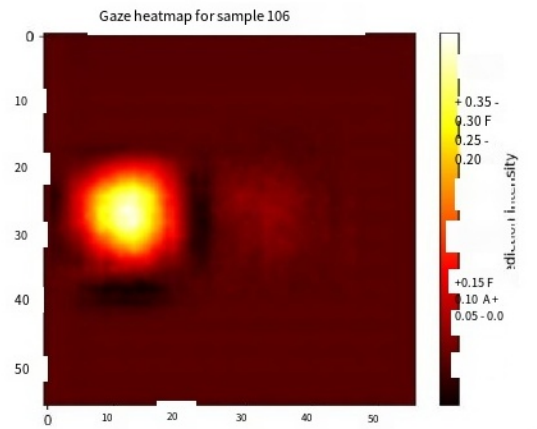
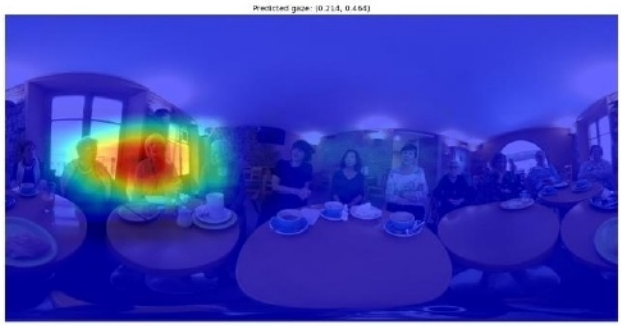
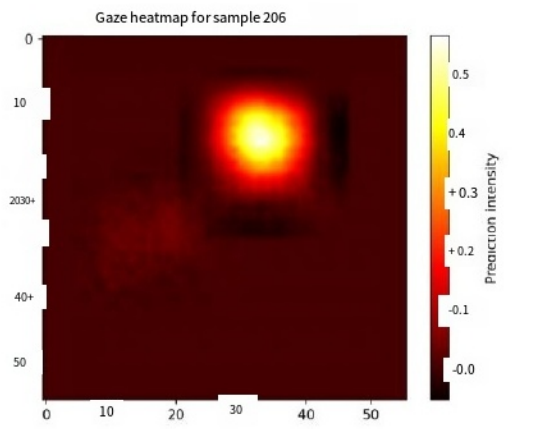
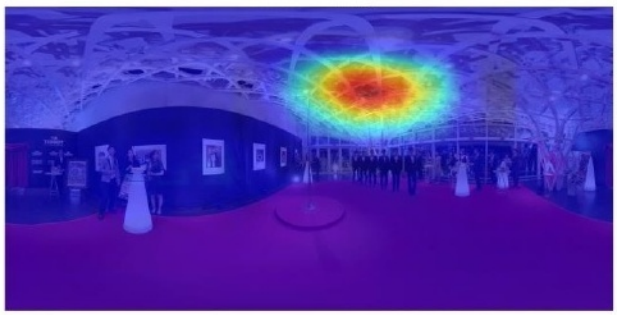
④ AUC: Area under the ROC curve, which measures the heatmap's ability to distinguish fixation locations; the closer the value is to 1, the better.

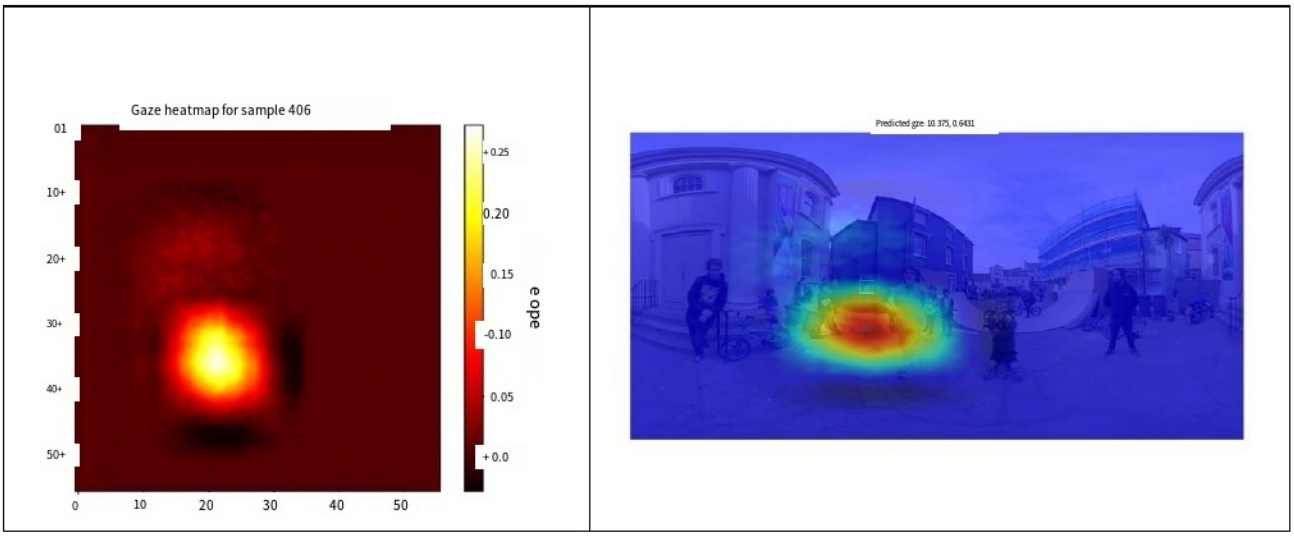
The quantitative results of the final model on the test set are shown in Figure 4-1. The normalized distance is 0.2426, the spherical distance is 0.99 radians, and the AUC is 0.7129. Due to hardware limitations, we achieved an average accuracy of 71% by training only five epochs at each stage, which validates the effectiveness of the model.

```
PS D:\something_of_JNU\2025dachuang\gazefollow360-main\gazefollow360-main> & D:\anaconda\envs\gaze360\python.exe eval_metrics.py
100%|                                                
```

测。

Table 4-1 Partial fusion attention heatmap results display



4.2 Results Comparison and Analysis

4.2.1 3D Coordinate Reconstruction

In tests evaluating the impact of wide-angle lens distortion on coordinate transformation accuracy, the project team found that when using uncorrected images for coordinate transformation, reconstructed 3D points in edge regions exhibited significant systematic deviations, with non-physical distortions observed in certain areas and transformation errors reaching tens of pixels at image edges. After correcting with distortion parameters output by COLMAP, the geometric consistency of point clouds was significantly improved, and errors in edge regions were substantially reduced, providing a reliable foundation for precise spatial mapping of line-of-sight vectors.

Compared to the initial three-layer coordinate transformation approach based on 360-degree panoramic completion, the current COLMAP-based multi-layer transformation method performs better in the following aspects: pseudo-information introduced by image inpainting regions in panoramic completion can interfere with feature matching, whereas the COLMAP approach directly uses real multi-view images, achieving higher reconstruction accuracy; spherical coordinate chains involve greater computational load than direct matrix operations, while the current method modularizes computations across layers through a well-designed hierarchy, enabling efficient execution during online inference.

Table 4-2 Comparison of Two 3D Coordinate Modeling Schemes

Solution	Applicable scenarios	Geometric accuracy	Computational efficiency	Deployment difficulty
Panoramic Completion with Three-Stage Coordinate Transformation (Initial)	Only 360° panoramic camera	Middle (edge area)	China	Low
Period		Large error		
COLMAP + Multi-layer Coordinate Transformation (current)	Single or multiple cameras	High (distortion correction)	High (modular design)	China
Line		After	Count	

4.2.2 Eye estimation

In the comparative experiments on model architecture selection, the project team evaluated the performance of different ResNet variants on the MPIIFaceGaze dataset. ResNet-50 achieved a good balance between accuracy and speed as the baseline architecture: compared to shallower networks such as ResNet-18/34, its Bottleneck structure provided stronger feature representation capability, resulting in significantly lower MAE; compared to deeper networks like ResNet-101/152, ResNet-50 showed only limited improvement in gaze estimation accuracy but offered noticeably faster inference speed.

Fast.

In the ablation study on loss function design, the hybrid loss function (cross-entropy + α·MSE) outperforms single losses. Pure cross-entropy loss can only output discrete angle predictions at bucket levels, with resolution limited by bucket width; pure MSE regression struggles to converge over large angular ranges; whereas the hybrid approach combines the stability of classification with the fine-grained precision of regression, achieving the best MAE on both datasets.

Table 4-3 Comparison of Different Loss Function Strategies

Comparison Dimensions	Pure Classification (CE)	Pure Regression (MSE)	Hybrid strategy (CE + α·MSE)
Angular resolution	Limited by bucket width	Continuous	Continuous
Training stability	High	Low (wide angle range)	High
Prediction accuracy	China	Low to medium	High
L2CS-Net adopts	×	×	√

4.2.3 Comparison with traditional gazefollow360 model for gaze point prediction

To validate the effectiveness of the training, we designed ablation experiments to compare the gaze prediction performance of the gazefollow360 model with 3D reconstruction and the improved L2CS-Net against the original gazefollow360 model under both uniform training epochs and a staged progressive training approach. The experimental results are shown in Table 4-4. The improved model outperforms the traditional gazefollow360 model across all evaluation metrics.

Table 4-4 Comparison of Two 3D Coordinate Modeling Schemes

Solution	Normalized distance	Spherical distance	AUC
Traditional gaze follow 360	0.354	1.23	0.68
3D Reconstruction + Improved L2CS_Netgazefollow360	0.243	0.99	0.71

4.3 Application Effect Evaluation and Discussion

4.3.1 3D Coordinate Reconstruction

In the integration testing with the overall system, the coordinate transformation results output by this module effectively support the judgment of gaze-target matching. Taking a single-camera laboratory scenario as an example, the facial center coordinates output by RetinaFace are processed through the coordinate transformation chain to obtain the gaze direction vector in the subject's coordinate system. This vector is then compared with the boundary point coordinate range of the cup reconstructed by COLMAP; when the gaze vector falls within the target boundary range, it is identified as valid fixation, and the counting logic operates correctly.

In the theoretical validation of multi-camera scenarios, the line-of-sight vectors from different camera coordinate systems were successfully unified into the world coordinate system using the rotation matrices provided by COLMAP, thereby establishing a technical pathway for subsequent multi-camera collaborative analysis.

The main limitations of the current approach are: 3D reconstruction is performed offline, requiring re-construction whenever the scene layout changes; SfM demands high-quality input images, and reconstruction quality degrades under frequent dynamic occlusions or insufficient image overlap; and the full pipeline running entirely on the CPU.

Shi Xing still experiences pressure when dealing with a large number of people, and further optimization can be achieved later by integrating model quantization and inference acceleration engines.

4.3.2 Eye estimation

In the integration testing with the overall security system, the gaze direction prediction results output by this module effectively support downstream tasks. Taking a single-camera laboratory scenario as an example, RetinaFace successfully detects all visible faces in the frame (with confidence threshold above 0.5), and L2CS-Net outputs yaw and pitch angle predictions for each detected face. Using visualization tools, gaze arrows are drawn on the image, and the indicated directions subjectively align well with the actual gaze directions of the individuals.

In real-time testing, the pipeline achieves end-to-end processing of single-frame multi-face images within 50 ms on GPU (CUDA), including RetinaFace detection and L2CS-Net prediction, meeting the real-time requirement of over 20 fps. Although inference speed decreases on CPU, it still satisfies the requirements for offline video analysis.

The main limitations of the current approach are as follows: the model's prediction accuracy significantly decreases at extreme deflection angles (close to ±180°), due to sparse sample distribution in the training data for these extreme angle ranges; under degraded conditions such as low illumination and severe occlusion, RetinaFace's detection accuracy drops, indirectly affecting gaze prediction performance; furthermore, the current model processes only single static frames without leveraging temporal continuity information from videos, resulting in frame-to-frame prediction jitter.



Figure 4-9 Line-of-sight estimation prediction map

4.3.3 Gaze prediction

The performance of this model in real-world application scenarios can be evaluated from the following aspects:

Robustness: The test set includes samples under varying lighting, poses, and occlusion conditions. The model maintains good performance even when the face is partially occluded (e.g., wearing glasses or a mask), and when the face is rotated at large angles (side view exceeding 60°), it employs 3D facial landmarks.

Key point refinement can effectively improve the model's prediction performance.

Interpretability: By visualizing intermediate heatmaps, the model's decision-making rationale can be explained. For instance, when predicting a gaze point on a distant building, the far-distance heatmap will show high response in that region, while the near-distance heatmap may focus on distractors near the head. This aids in debugging incorrect predictions: if neither heatmap is focused, it may indicate an error in direction estimation; if only the near-distance heatmap is incorrect, it could suggest improper region cropping.

Continuity: Explicit 3D modeling can effectively enhance the boundary continuity of 360° panoramas, reducing gaze prediction errors near the edges.

Limitation: The model assumes only one person per image; for multi-person scenes, person detection must be performed first.

4.4 Summary of Achievements

(1) 3D Coordinate Reconstruction

This module successfully implemented a spatial modeling system based on COLMAP 3D reconstruction and a multi-layer coordinate transformation framework. Key achievements include: (1) complete code implementation and testing of a four-stage progressive transformation chain from image plane coordinates → spherical coordinates → camera coordinates → object coordinates; (2) successful 3D reconstruction of laboratory scenes using COLMAP, obtaining accurate camera parameters and sparse point clouds, and accurately calibrating the 3D boundary coordinates of target objects; (3) validation of the wide-angle lens distortion correction method, significantly improving coordinate transformation accuracy in edge regions; (4) theoretical verification of the multi-camera coordinate unification approach, demonstrating practical deployment feasibility.

The above results provide a reliable three-dimensional spatial foundation for the gaze target identification module in the overall security early warning system, directly supporting the precise matching of line-of-sight vectors with sensitive targets.

(2) Gaze Estimation

This module successfully implemented a complete gaze direction prediction system based on the L2CS-Net architecture, with key achievements including: (1) implementing and training a ResNet-50 backbone network combined with a dual-head classification-regression joint prediction model, validated on two benchmark datasets—Gaze360 and MPIIFaceGaze; (2) designing and implementing an end-to-end inference pipeline integrating RetinaFace face detection with L2CS-Net gaze estimation, supporting various input sources such as real-time cameras and video files; (3) achieving a best MAE of 7.23° on the fold0 test set in the MPIIFaceGaze 15-fold cross-validation, with overall MAE consistently ranging between 7° and 11° across all folds, demonstrating strong inter-subject generalization capability; (4) developing a gaze direction visualization tool capable of drawing 3D gaze arrows and facial bounding boxes directly onto images.

The above results provide reliable gaze direction input for the target fixation judgment module in the overall security early warning system, directly supporting the spatial matching analysis from gaze vectors to fixation targets.

(3) Overall Application Value

This project successfully developed a 360-degree panoramic image gaze prediction model based on progressive deep learning, achieving the following practical applications: the model can be applied to eye-tracking interaction in virtual reality, driver attention monitoring, and customer interest analysis in product displays. The provided real-time prediction code can be directly integrated into existing systems.

4.5 Shortcomings and Prospects

(1) 3D Coordinate Reconstruction

The current shortcomings of this module are mainly reflected in three aspects: first, the offline nature of 3D reconstruction limits its adaptability to dynamic scene changes; second, relying solely on 3D point clouds for gaze target matching leads to ambiguity in crowded and frequently occluded scenes; third, the real-time performance of the current coordinate transformation chain in a CPU environment requires further validation and optimization.

Subsequent work will proceed in the following directions: integrating scene semantic segmentation to extend gaze-mapping targets from 3D point clouds to semantic regions, thereby improving target recognition accuracy in dense scenes; introducing temporal smoothing filtering algorithms to suppress gaze estimation jitter and distinguish intentional fixation from involuntary saccades; exploring inference acceleration solutions using TensorRT/ OpenVINO to further reduce end-to-end pipeline latency and meet performance requirements for real-time, concurrent warnings across multiple users.

(2) Gaze Estimation

The current shortcomings of this module are evident in four aspects: first, the model's prediction accuracy at extreme angles needs improvement, especially in near ±180° back-facing gaze scenarios; second, the current training strategy and data augmentation scheme have not yet fully aligned with the optimal configuration described in the paper, resulting in a certain gap compared to the original L2CS-Net report results; third, the model relies solely on single-frame information for prediction without incorporating temporal modeling mechanisms, leading to jitter in inter-frame predictions; fourth, the ResNet-50 model size (approximately 25.6 million parameters) still has room for compression in resource-constrained edge deployment scenarios.

Subsequent work will proceed in the following directions: replacing the ResNet-50 backbone with EfficientNet-B4 to significantly reduce model parameters and inference latency while maintaining prediction accuracy, thereby meeting deployment requirements on edge devices; integrating the Temporal Shift Module (TSM) for temporal modeling to leverage sequential information across video frames, smoothing prediction jitter and distinguishing between intentional gaze and involuntary saccadic movements; optimizing training strategies by incorporating data augmentation techniques such as random flipping, color jittering, and Mixup, extending the training duration, and employing cosine annealing learning rate scheduling to further approach the optimal accuracy reported in the paper; exploring model quantization (INT8/FP16) and TensorRT-based inference acceleration to further compress end-to-end pipeline latency, satisfying performance requirements for real-time, concurrent early warning of multiple individuals.

(3) Fixation Prediction Evaluation

Although this model achieves promising performance, several limitations and areas for improvement remain: ① Limited dataset size: GazeFollow360 contains only about 9,000 samples with insufficient scene diversity (mostly outdoor performances and indoor interviews). The model's generalization capability on unseen scene types (e.g., streets, stores) needs further validation. ② Single-person assumption: the model processes only one person per image. For multi-person gaze interaction scenarios, additional modules for person detection and relationship reasoning are required.

Subsequent work will proceed in the following directions: ① Data augmentation and collection: expand the dataset using synthetic data (e.g., rendering 360° images of various scenes with Unity) and annotate more diverse gaze behaviors. Utilize semi-supervised learning to leverage unlabeled data; ② Multi-person gaze prediction: extend the model to a multi-person version by first detecting all faces, then independently predicting gaze points for each individual, while incorporating inter-personal attention mechanisms to model social gaze interactions; ③ Incorporating temporal information: employ ConvLSTM or Transformer to model features across consecutive frames temporally, improving prediction stability and enabling gaze trajectory forecasting.

With the above improvements, this model is expected to play a role in a wider range of real-world scenarios, providing more intelligent gaze perception capabilities for fields such as human-computer interaction and virtual reality.

5. Innovation Point

This project addresses the issue of gaze behavior analysis in 360-degree panoramic monitoring scenarios, with improvements made in three-dimensional space modeling, coordinate transformation mechanisms, and system implementation.

(1) Constructing a Three-Tier Progressive Coordinate Transformation Framework for Panoramic Monitoring

Most existing eye-tracking studies are limited to two-dimensional image planes or simple three-dimensional camera coordinate systems, making them unsuitable for handling spherical projection distortions in 360-degree panoramic surveillance scenarios. Additionally, they lack modeling of the viewer's subjective perspective, resulting in poor generalization across different scenes. To address these issues, this project proposes a three-tier progressive coordinate transformation framework consisting of "image plane coordinates → spherical coordinates → camera coordinates → subject coordinates."

The framework starts with the image plane coordinate system, mapping 2D pixel coordinates to spherical coordinates via the inverse transformation of an equirectangular cylindrical projection, effectively addressing geometric distortions present in panoramic images. Subsequently, by converting spherical coordinates into Cartesian coordinates, the gaze direction is unified within the camera coordinate system, providing a convenient mathematical representation for subsequent spatial computations. Building upon this, the project draws inspiration from robotics' concept of body-fixed coordinate systems, introducing a subject-centered coordinate system with the center of the person's head as the origin. This decouples the representation of gaze direction from the individual's absolute position in the scene, ensuring intrinsic consistency and comparability in gaze descriptions. This design not only enhances the system's generalization capability across diverse scenarios but also lays a theoretical foundation for the semantic understanding of eye-gaze behaviors.

(2) Achieve precise mapping between the line of sight direction and the 3D scene

This project integrates multi-view geometry-based 3D reconstruction techniques, using COLMAP to perform offline reconstruction of surveillance scenes, obtaining accurate camera parameters and scene point cloud models, thereby providing a high-precision spatial reference frame for line-of-sight mapping.

In terms of mapping mechanisms, the project implements a complete transformation pipeline from the viewer's coordinate system to the world coordinate system, unifying the gaze vector into the reconstructed scene's coordinate frame through matrix operations. Building upon this, ray tracing algorithms are employed to determine the intersection points between the gaze vector and predefined target objects, while a temporal filtering mechanism suppresses jitter errors in single-frame predictions. This approach effectively addresses the challenge of aligning gaze direction with spatial targets in panoramic surveillance, enabling precise mapping from two-dimensional images to three-dimensional space.

(3) Developing a Lightweight Gaze Estimation Model and System Integration Solution

To meet the real-time requirements of practical monitoring scenarios, the project has implemented targeted optimizations in model design and system integration. At the gaze estimation model level, based on the L2CS-Net architecture, the project replaces the traditional ResNet-50 backbone with EfficientNet-B4, significantly reducing the number of parameters and computational complexity while maintaining high accuracy, thereby improving inference efficiency on CPU environments.

At the system integration level, the project established a complete processing pipeline covering video input, face detection, gaze estimation, and target fixation determination. Through modular design and standardized interfaces, efficient collaboration among modules was achieved. Experimental validation in laboratory settings demonstrated that the system can reliably output gaze direction and fixation targets for individual subjects, providing solid technical support for subsequent behavior analysis and early warning functions.

References

[1] 毛云丰, 沈文忠, 滕童. 基于深度神经网络的视线跟踪技术研究[J]. 现代电子技术,2020,43(16):39-43

[2] 胡琥.视线跟踪技术及其在人机交互中的应用.黑龙江东方学院.2011

[3] 吴艳繁. 基于视线追踪的人机交互系统[D]. 吉林大学[2025-03-08].DOI:CNKI:CDMD:2.1014.281995.

[4] Xuan-Bac Nguyen, Chi Nhan Duong, Xin Li, Susan Gauch, Han-Seok Seo, Khoa Luu. Micron-BERT: BERT-based Facial Micro-Expression Recognition. Retired from https://arxiv.org/abs/2304.03195v1

[5] Fernando Alonso-Fernandez, Kevin Hernandez-Diaz, Jose Maria Buades Rubio, Josef Bigun. SqueezerFaceNet: Reducing a Small Face Recognition CNN Even More Via Filter Pruning. Retired from https://doi.org/10.48550/arXiv.2307.10697

[6] Fromherz T , Stucki P , Bichsel M .A Survey of Face Recognition[J]. 1997.

[7] FU Youjia. Facial Pose Estimation Method on Single Image Based on Facial Feature Points[J]. Computer Engineering, 2021, 47(4): 197-203,210.

[8] MIN Qiusha,LIU Neng,CHEN Yating,WANG Zhifeng. Head Pose Estimation Based on Facial Feature Point Localization[J]. Computer Engineering, 2018, 44(6): 263-269.

[9] 马利庄,吴飞,毛启容,等.面向复杂场景的人物视觉理解技术 [J].中国图象图形学 报,2022,27(6):20

[10]万子敬.基于空间瞳孔的视线跟踪及注视点空间坐标测量方法研究.天津大学.2018

[11]苏松志，李绍滋，陈淑媛. 行人检测技术综述[J]. 电子学报，2012，40(4): 814 ~ 820.

[12]王长元,蒋光毅,贾宏博.视线追踪与 VR 融合交互方法研究[C]//中国自动化学会;中国计算 机用户协会.中国自动化学会;中国计算机用户协会,2016.

[13]王立松, 宋佳琦, 吕洋. 基于眼动跟踪技术诊断阿尔茨海默病的Logistics 回归模型的初步建立[J]. 陆军军医大学学报, 2023, 45(2): 102-110.